

Aadit Gupta

aadit.gupta@mail.utoronto.ca | +1 (437) 580-6452 | github.com/aaditguptaa | linkedin.com/in/aadit-gupta-ag

Education

University of Toronto

Sep. 2024 – Apr. 2029 (expected)

Bachelor of Applied Science and Engineering in Computer Engineering + PEY Co-op

Toronto, ON

- Relevant Courses: Calculus III; Computer Fundamentals (C/C++); Digital Systems (Verilog/FPGA Design)
- In Progress: Deep Learning; Probability; Intro to AI; Computer Architecture; Operating Systems; DSA
- Intended Minors & Certificate: AI Engineering Minor; Robotics & Mechatronics Minor; Engineering Business Certificate

Technical and Professional Skills

Coding Languages: Python, C/C++, JavaScript/TypeScript, SQL, Verilog (HDL)

Machine Learning & Data: PyTorch, Pandas, OpenAI API, NSGA-II/pymoo, ChromaDB, MySQL

Frameworks & Libraries: React, Node.js, Django, Flask, FastAPI, OpenCV

Tools & Platforms: Git, Docker, Linux, MATLAB, LTSpice, Altium, Arduino IDE

Professional Skills: Teamwork, Leadership, Easily Adaptable, Technical Documentation

Work Experience

Machine Learning Researcher | University of Strathclyde

Jun. 2026 – Aug. 2026

github.com/aaditguptaa/CAELUS_Project

Glasgow, UK

- Built an **end-to-end digital twin** that evaluates in real time whether an AED-delivery drone can safely beat an ambulance to a cardiac-arrest call across Greater Glasgow & Clyde.
- Engineered an **A* drone routing algorithm** over population and wind-risk cost grids, tuning weights through parameter sweeps and **Pareto analysis** to minimize flight time while reducing population exposure.
- Developed a **PyTorch neural-network** surrogate for drone flight time prediction using a dataset of 20000 scenarios, with an 80/20 training-validation split and checkpoint selection based on validation error.
- Optimized drone-base placement with a multi-objective **NSGA-II genetic algorithm** (pymoo), raising reachable cardiac-arrest demand from **47% to 72%**.
- Implemented a JARUS SORA 2.5 **aviation-risk engine** with a **per-flight go/no-go engine** which was displayed using a **Flask** operator console featuring animated fleet simulation and coverage analytics.

AI Engineering Intern (Product Management Team) | Andersen UAE

Jun. 2025 – Aug. 2025

Dubai, UAE

- Developed an **AI E-Invoicing Assistant** that processed **tax queries in under 5 seconds**.
- Built and deployed a **RESTful API** with **FastAPI** serving RAG-based query endpoints, enabling client applications to retrieve context-aware responses from tax documents.
- Cleaned and Analyzed **10000+ data entries** using **Pandas** and stored vector representations in a **ChromaDB Database**.
- Reduced response times by **40%** through optimized database retrieval and **query-result caching**.
- Integrated GPT-4 via the **OpenAI API** into the pipeline to synthesize context into accurate, user-facing responses.

Co-Curricular Activities & Projects

University of Toronto Autonomous Scale Racing Team | Hardware Team Member

Sep. 2025 - Present

- Implemented **C++ UART** communication between an Arduino and an NVIDIA Jetson.
- Designed **PCB** circuitry in Altium to support communication between the Arduino Nano and the Jetson.
- Optimized communication by addressing timing issues and data size constraints on the Jetson.

University of Toronto Aerospace Team | Software Team Member

Sep. 2025 - Present

- Developed a **camera-based target localization system** using **Python** and **OpenCV** for autonomous missions.
- Implemented **minimum-jerk trajectory planning** in **MATLAB/Python**, focusing on smooth and stable flight paths.
- Containerized workflows using **Docker** to ensure reproducible development environments and consistent testing.

Streetview Monopoly Game Project | Node.js, MySQL

Jan. 2026 – Mar. 2026

- Built a **full-stack multiplayer game** leveraging **Google Streetview** for location-based gameplay.
- Developed server-side logic in **Node.js** for turn-based state management, player actions, and game rule enforcement.
- Designed a **MySQL database** schema to persist user accounts, game state and match history.

FPGA Design Project | Verilog, DE1-SoC, ModelSim

Sep. 2025 – Nov. 2025

- Designed a rhythm game using the **DE1-SoC Board**, which incorporated a VGA Display, Keyboard and Speakers.
- Implemented FSMs, Shift Registers and Counters in **Verilog** to implement game logic and output graphics on the VGA.
- Used **ModelSim** for **RTL simulation** and waveform analysis to validate module behavior and isolate timing errors.